COMPUTER VISION — COMPLETE REVISION GUIDE

UNIT 1 — FEATURE EXTRACTION

What is Feature Extraction?

Feature extraction is the process of finding important, distinctive information from an image that can be used for:

- object detection
- object recognition
- image matching
- tracking
- segmentation
- classification

Instead of processing every pixel, we extract useful features such as:

- **Edges** → boundaries
- **Corners** → distinctive points
- **Lines** → geometric structures
- **Texture** → repeated patterns
- **Local descriptors** → SIFT, SURF, ORB-like representations

Example:

```
Original image
      ↓
Edge detection
      ↓
Corners / keypoints
      ↓
Descriptors
      ↓
Object matching / recognition
```

1. EDGE DETECTION

An **edge** is a location in an image where there is a sudden change in intensity.

Mathematically:

$$|\nabla I(x,y)|$$

is large at an edge.

The gradient is:

$$\nabla I = \begin{bmatrix} \frac{\partial I}{\partial x} \\ \frac{\partial I}{\partial y} \end{bmatrix}$$

and gradient magnitude:

$$|\nabla I| = \sqrt{I_x^2 + I_y^2}$$

Gradient direction:

$$\theta = \tan^{-1}\left(\frac{I_y}{I_x}\right)$$

Applications

- object boundary detection
- lane detection
- shape detection
- segmentation
- OCR
- autonomous driving

2. CANNY EDGE DETECTOR

Canny is a **multi-stage edge detection algorithm.**

Five important steps

```
Image
 ↓
Gaussian smoothing
 ↓
Gradient calculation
 ↓
Non-Maximum Suppression
 ↓
Double Threshold
 ↓
Edge Tracking by Hysteresis
 ↓
Final edges
```

Step 1 — Gaussian smoothing

Removes noise.

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Smoothed image:

$$I_s = G * I$$

Step 2 — Gradient

Calculate:

$$G_x, \quad G_y$$

Magnitude:

$$G = \sqrt{G_x^2 + G_y^2}$$

Direction:

$$\theta = \tan^{-1} \frac{G_y}{G_x}$$

Step 3 — Non-Maximum Suppression

Makes thick edges **thin**.

Only retain a pixel if it is a local maximum along the gradient direction.

Step 4 — Double threshold

Two thresholds:

- High threshold → strong edge
- Low threshold → weak edge

Step 5 — Hysteresis

Weak edges connected to strong edges are retained.

Weak edges not connected to strong edges are removed.

Why Canny is popular?

It gives:

- good detection
- good localization
- thin edges
- relatively low noise sensitivity

3. LoG — Laplacian of Gaussian

LoG combines:

1. Gaussian smoothing
2. Laplacian operator

First smooth:

$$I_s = G_\sigma * I$$

Then:

$$LoG = \nabla^2 (G_\sigma * I)$$

where

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

An edge can be detected by looking for **zero crossings** of LoG.

Important idea

Canny → gradient based

LoG → second derivative based

Use

- edge detection
- blob detection
- multi-scale feature detection

4. DoG — Difference of Gaussian

DoG approximates LoG.

Take two Gaussian-blurred images with different scales:

$$DoG(x, y, \sigma) = G(x, y, k\sigma) - G(x, y, \sigma)$$

Then:

$$DoG * I$$

is calculated.

Why useful?

DoG is computationally cheaper than directly calculating LoG.

Major application

SIFT uses DoG for scale-space extrema detection.

5. Hough Transform — LINE DETECTION

Hough Transform detects shapes represented mathematically.

For a line:

$$y = mx + c$$

But the standard Hough representation is:

$$\rho = x\cos\theta + y\sin\theta$$

where:

- ρ = perpendicular distance from origin
- θ = angle of normal

Working

Each edge pixel votes for possible lines.

```
Edge pixels
     ↓
ρ, θ parameter space
     ↓
Accumulator voting
     ↓
Peaks
     ↓
Detected lines
```

Example

If many points satisfy:

$$\rho = 50, \quad \theta = 30^\circ$$

then the accumulator has a strong peak at that location.

Applications

- lane detection
- road boundaries
- document borders
- industrial inspection
- detecting geometric shapes

6. CORNER DETECTION

A corner is a point where image intensity changes significantly in **multiple directions**.

Think:



Corners are useful because they are usually distinctive.

7. HARRIS CORNER DETECTOR

Harris examines how intensity changes when a small window is shifted.

Structure matrix:

$$M = \begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix}$$

Harris response:

$$R = \det(M) - k(\text{trace}(M))^2$$

Since:

$$\det(M) = \lambda_1 \lambda_2$$

and

$$trace(M) = \lambda_1 + \lambda_2$$

we have:

$$R = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$$

Interpretation

λ_1	λ_2	Result
small	small	Flat
large	small	Edge
large	large	Corner

Important exam point

Harris is **rotation invariant**, but not inherently scale invariant.

8. HESSIAN-AFFINE

Hessian matrix:

$$H = \begin{bmatrix} I_{xx} & I_{xy} \\ I_{xy} & I_{yy} \end{bmatrix}$$

Determinant:

$$\det(H) = I_{xx} I_{yy} - I_{xy}^2$$

It identifies blob-like structures and uses affine adaptation to obtain features robust to viewpoint changes.

Applications

- image matching
- object recognition
- wide-baseline matching

9. ORIENTATION HISTOGRAM

An orientation histogram represents how frequently different gradient directions occur in an image region.

For each pixel:

$$m(x,y) = \sqrt{I_x^2 + I_y^2}$$

$$\theta(x,y) = \tan^{-1} \frac{I_y}{I_x}$$

Then magnitudes vote into orientation bins.

Example:



This concept is fundamental to **SIFT and HOG**.

10. SIFT

Scale-Invariant Feature Transform

SIFT generates keypoints and descriptors that are relatively robust to:

- scale
- rotation
- illumination changes
- moderate viewpoint changes

Four major stages

```
1. Scale-space extrema detection
        ↓
2. Keypoint localization
        ↓
3. Orientation assignment
        ↓
4. Descriptor generation
```

Step 1 — Scale space

Construct Gaussian blurred images at different scales.

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

DoG:

$$D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma)$$

Find local extrema.

Step 2 — Keypoint localization

Remove unstable points and low-contrast points.

Step 3 — Orientation

Calculate gradient magnitude and orientation.

$$m = \sqrt{L_x^2 + L_y^2}$$

$$\theta = \tan^{-1} \frac{L_y}{L_x}$$

Step 4 — Descriptor

A typical SIFT descriptor has:

$$4 \times 4 \times 8 = 128$$

dimensions.

Applications

- panorama stitching
- image matching
- object recognition
- image retrieval

11. SURF

Speeded-Up Robust Features

SURF was designed to be faster than SIFT.

Uses:

- Hessian-based detector
- integral images
- Haar wavelets

Typical SURF descriptor:

$$64$$

dimensions, although extended variants can use 128.

SIFT vs SURF

SIFT	SURF
DoG	Hessian
128-D descriptor	Usually 64-D
More computationally expensive	Faster
Gradient histogram	Haar-wavelet responses

12. HOG

Histogram of Oriented Gradients

HOG describes an object using local gradient orientations.

Steps

```
Image
 ↓
Gradient calculation
 ↓
Orientation histogram
 ↓
Cell normalization
 ↓
Block normalization
 ↓
HOG descriptor
```

Gradient:

$$G_x = I(x+1, y) - I(x-1, y)$$

$$G_y = I(x, y+1) - I(x, y-1)$$

Magnitude:

$$M = \sqrt{G_x^2 + G_y^2}$$

Orientation:

$$\theta = \tan^{-1} \frac{G_y}{G_x}$$

Famous application

Pedestrian detection.

13. GLOH

Gradient Location and Orientation Histogram

GLOH extends SIFT by using a **log-polar spatial arrangement**.

It captures:

- gradient orientation
- gradient location

It can produce a high-dimensional descriptor which is commonly reduced using **PCA**.

SIFT vs GLOH

SIFT:

> rectangular 4×4 spatial regions

GLOH:

> log-polar spatial regions

14. SCALE-SPACE ANALYSIS

Objects can appear at different sizes.

Therefore, computer vision needs features that work across multiple scales.

Scale-space representation:

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

where σ controls scale.

Small σ:

> fine details

Large σ:

> coarse structures

15. IMAGE PYRAMIDS

An image pyramid represents an image at multiple resolutions.

Gaussian pyramid

```
Original
   ↓
Blur + Downsample
   ↓
Level 1
   ↓
Blur + Downsample
   ↓
Level 2
```

Applications:

- multi-scale object detection
- image blending
- optical flow
- image registration

Laplacian pyramid

Represents differences between Gaussian pyramid levels.

Useful for:

- image compression
- image blending
- multi-resolution analysis

16. GAUSSIAN DERIVATIVE FILTERS

Instead of differentiating a noisy image directly, first smooth with Gaussian.

First derivative:

$$\frac{\partial G}{\partial x}$$

Second derivative:

$$\frac{\partial^2 G}{\partial x^2}$$

This reduces noise sensitivity.

Applications:

- edge detection
- blob detection
- feature extraction

17. GABOR FILTER

Gabor filter is a sinusoidal wave modulated by a Gaussian.

General form:

$$G(x,y) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right)\cos\left(2\pi\frac{x'}{\lambda} + \phi\right)$$

where rotated coordinates are:

$$x' = x\cos\theta + y\sin\theta$$

$$y' = -x\sin\theta + y\cos\theta$$

It is particularly useful for **texture analysis**.

Applications:

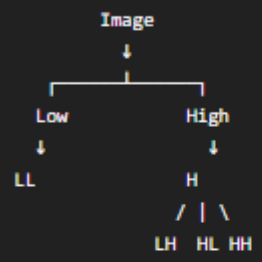
- texture segmentation
- fingerprint analysis
- face recognition
- texture classification

18. DWT

Discrete Wavelet Transform

DWT decomposes an image into frequency components.

For a single level:



Actually, the four subbands are:

- LL → low-low
- LH → low-high
- HL → high-low
- HH → high-high

LL

Contains coarse approximation.

High-frequency bands

Contain:

- edges
- details
- textures

Applications:

- compression
- denoising
- texture analysis
- watermarking

UNIT 2 — IMAGE SEGMENTATION

What is Image Segmentation?

Segmentation divides an image into meaningful regions.

Example:



Goal:

> Group pixels belonging to the same object/region.

19. REGION GROWING

Start with one or more **seed pixels.**

Add neighboring pixels if they satisfy a similarity condition.

Example condition:

$$|I(p) - I(seed)| < T$$

Algorithm

```
Choose seed
 ↓
Check neighbors
 ↓
Similar?
 ↓ YES
Add to region
 ↓
Repeat
```

Advantages

- simple
- produces connected regions

Disadvantages

- seed dependent
- threshold dependent
- sensitive to noise

20. EDGE-BASED SEGMENTATION

Idea:

> Find boundaries first, then use boundaries to separate objects.

Typical pipeline:

$$Image \rightarrow Edge\ Detection \rightarrow Boundary\ Formation \rightarrow Segmentation$$

Canny is commonly used.

Problem

Broken/noisy edges can cause poor segmentation.

21. GRAPH CUT

Represent the image as a graph.

- pixels → nodes
- neighboring pixels → edges
- source → foreground
- sink → background

Then segmentation is obtained by finding a minimum cut.

Energy formulation:

$$E(L) = \sum_{p} D_p(L_p) + \lambda \sum_{(p,q)} V_{pq}(L_p, L_q)$$

where:

- D_p = data cost
- V_{pq} = smoothness cost
- λ = weighting parameter

Goal:

$$\boxed{\min E(L)}$$

Applications

- foreground/background segmentation
- medical image segmentation
- interactive segmentation

22. MEAN-SHIFT SEGMENTATION

Mean-shift is a **non-parametric clustering technique.**

It moves a point toward the region of highest data density.

Basic idea:

```
Points
 ↓
Calculate local mean
 ↓
Move toward mean
 ↓
Repeat
 ↓
Converged modes
```

Applications:

- image segmentation
- color clustering
- object tracking

Important advantage

Doesn't require the number of clusters K beforehand.

23. MRF — MARKOV RANDOM FIELD

MRF models relationships between neighboring pixels.

The fundamental assumption:

> The label of a pixel depends mainly on neighboring labels.

Energy:

$$E(X) = \sum_i D_i(X_i) + \sum_{i,j} V_{ij}(X_i, X_j)$$

Again:

- unary/data term
- pairwise/smoothness term

Applications

- image denoising
- segmentation
- stereo
- labeling problems

24. TEXTURE SEGMENTATION

Texture refers to patterns such as:

- grass
- brick
- fabric
- wood
- sand

Texture segmentation separates regions based on texture rather than simply intensity.

Methods:

- Gabor filters
- wavelets/DWT
- Local Binary Patterns
- Laws' filters
- statistical texture features

25. OBJECT DETECTION

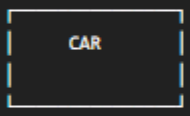
Object detection answers:

> **What object is present and where is it?**

Output usually contains:

- object class
- bounding box
- confidence

Example:



Traditional CV pipeline:

$$Image \rightarrow Features \rightarrow Classifier \rightarrow Object$$

Examples:

- HOG + SVM → pedestrian detection
- Haar features + cascade → face detection

Modern systems include:

- YOLO
- SSD
- Faster R-CNN

UNIT 3 — PATTERN ANALYSIS

26. PATTERN ANALYSIS

Pattern analysis means finding structure in data and assigning observations to meaningful groups/classes.

Two major tasks:

Clustering

No labels.

Classification

Known labels.

27. K-MEANS

K-Means partitions data into K clusters.

Algorithm

1. Choose K
2. Initialize centroids
3. Assign each point to nearest centroid
4. Recalculate centroids
5. Repeat

Distance:

$$d(x, \mu) = \|x - \mu\|^2$$

Objective:

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

Numerical example

Data:

$$1, 2, 8, 9$$

Take $K = 2$.

Initial centroids:

$$\mu_1 = 1, \quad \mu_2 = 8$$

Assignment:

$$1, 2 \to C_1$$

$$8, 9 \to C_2$$

New centroids:

$$\mu_1 = \frac{1+2}{2} = 1.5$$

$$\mu_2 = \frac{8+9}{2} = 8.5$$

Repeat until stable.

28. K-MEDOIDS

Similar to K-Means, but:

K-Means

Centroid may be an artificial point.

K-Medoids

Center must be an **actual data point.**

This makes K-Medoids more robust to outliers.

Example:

```
Data: 1, 2, 3, 100

K-Means center = 26.5

K-Medoids center = one actual point
```

29. MIXTURE OF GAUSSIANS

Gaussian Mixture Model assumes data comes from multiple Gaussian distributions.

$$p(x) = \sum_{k=1}^{K} \pi_k N(x|\mu_k, \Sigma_k)$$

where:

- π_k = mixing coefficient
- μ_k = mean
- Σ_k = covariance

Constraints:

$$\sum_k \pi_k = 1$$

Usually trained using **Expectation-Maximization (EM)**.

K-Means vs GMM

K-Means:

> hard assignment

GMM:

> probabilistic/soft assignment

30. CLASSIFICATION

Classification assigns an input to a predefined class.

Example:

```
Image
 ↓
Feature extraction
 ↓
Classifier
 ↓
Car / Bike / Person
```

---

31. DISCRIMINANT FUNCTION

A discriminant function assigns a score to each class.

$$g_i(x)$$

Choose:

$$\boxed{\text{Class} = \arg\max_i g_i(x)}$$

For a linear classifier:

$$g(x) = w^T x + w_0$$

Decision boundary:

$$w^T x + w_0 = 0$$

---

32. SUPERVISED LEARNING

Training data has labels.

```
Image → Feature → Label
Car   → feature → Car
Bike  → feature → Bike
```

Examples:

- SVM
- KNN
- neural networks
- Bayes classifier

---

33. UNSUPERVISED LEARNING

No labels.

Algorithm discovers structure.

Examples:

- K-Means
- K-Medoids
- GMM
- Mean-shift

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34. SEMI-SUPERVISED LEARNING

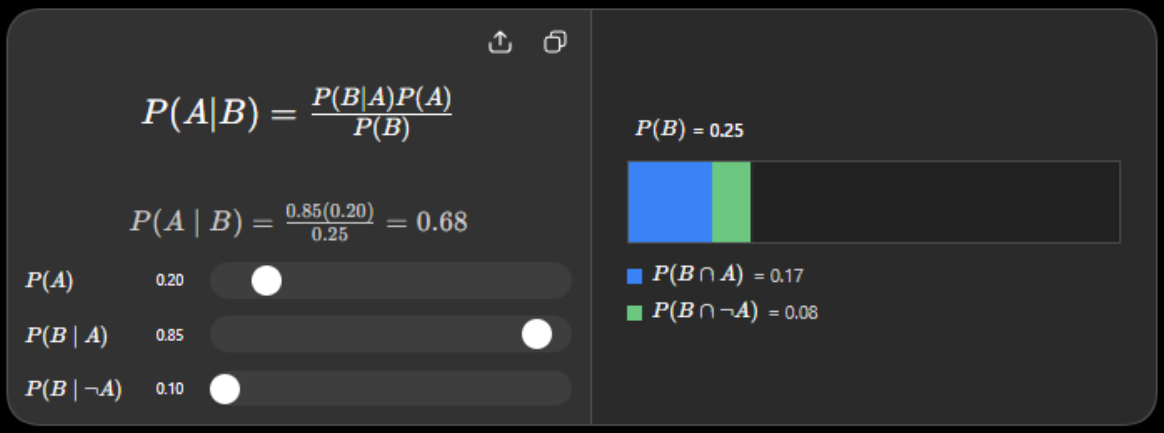
Combination:

```
Small labeled dataset
        +
Large unlabeled dataset
        ↓
       Model
```

Useful when labeling images is expensive.

35. BAYES CLASSIFIER

Bayes theorem:



Give feedback

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

where:

- $P(C|X)$ = posterior
- $P(X|C)$ = likelihood
- $P(C)$ = prior
- $P(X)$ = evidence

Choose:

$$\boxed{\arg\max_C P(C|X)}$$

Numerical example

Suppose:

$$P(Car) = 0.6$$

$$P(Bike) = 0.4$$

and:

$$P(X|Car) = 0.8$$

$$P(X|Bike) = 0.3$$

Then unnormalized posterior:

$$Car = 0.8(0.6) = 0.48$$

$$Bike = 0.3(0.4) = 0.12$$

Therefore the Bayes classifier assigns X to the class with the larger posterior.

36. KNN

K-Nearest Neighbors

No explicit training model is built.

For a test point:

1. Calculate distances
2. Find K nearest points
3. Majority vote

Euclidean distance:

$$d(x,y) = \sqrt{\sum_i (x_i - y_i)^2}$$

Example

For $K = 3$:

```
Nearest:
Car
Car
Bike
```

Prediction:

$$\boxed{Car}$$

Important

Small K:

> sensitive to noise

Large K:

> smoother decision boundary

37. ANN

Artificial Neural Network

Basic neuron:

$$z = \sum_i w_i x_i + b$$

Activation:

$$a = f(z)$$

Common activation:

Sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

ReLU

$$ReLU(x) = \max(0, x)$$

A neural network consists of:

```
Input layer
     ↓
Hidden layer(s)
     ↓
Output layer
```

Training generally uses:

- forward propagation
- loss calculation
- backpropagation
- gradient descent

UNIT 4 — DIMENSIONALITY REDUCTION

Why?

Suppose an image feature has:

$$10,000$$

dimensions.

Processing all dimensions can be expensive.

Dimensionality reduction converts:

$$10,000 \rightarrow 100$$

while trying to retain important information.

38. PCA

Principal Component Analysis

PCA finds directions of maximum variance.

Steps

1. Center data
2. Calculate covariance matrix
3. Find eigenvalues/eigenvectors
4. Sort eigenvectors by eigenvalue
5. Select top k
6. Project data

Mean:

$$\mu = \frac{1}{n}\sum_i x_i$$

Centered data:

$$X_c = X - \mu$$

Covariance:

$$C = \frac{1}{n-1}X_c^T X_c$$

Eigenvalue equation:

$$Cv = \lambda v$$

Largest eigenvalues correspond to principal components.

Projection:

$$Z = X_c W$$

Key point

PCA is **unsupervised.**

It maximizes variance.

39. LDA

Linear Discriminant Analysis

Unlike PCA, LDA uses class labels.

Goal:

> maximize separation between classes while minimizing spread within each class.

Between-class scatter:

$$S_B$$

Within-class scatter:

$$S_W$$

Objective:

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

Maximize $J(w)$.

PCA vs LDA

PCA	LDA
Unsupervised	Supervised
Ignores class labels	Uses class labels
Maximum variance	Maximum class separation
Good compression	Good classification preprocessing

40. ICA

Independent Component Analysis

ICA tries to represent observed signals as statistically independent components.

Model:

$$X = AS$$

where:

- X = observed signals
- A = mixing matrix
- S = independent source signals

Goal:

$$S = A^{-1}X$$

Famous example

Cocktail party problem

Several people speak simultaneously, microphones record mixtures, ICA attempts to recover individual voices.

41. NON-PARAMETRIC METHODS

Parametric methods assume a fixed distribution.

Example:

$$X \sim N(\mu, \sigma^2)$$

Non-parametric methods don't require a specific distribution assumption.

Examples:

- KNN
- Kernel Density Estimation
- Parzen windows
- histogram methods
- decision trees

Advantages

Flexible.

Disadvantages

Usually require more data.

UNIT 5 — MOTION ANALYSIS

Now we move from **static images → video.**

Motion analysis determines how objects/pixels move between frames.

---

42. BACKGROUND SUBTRACTION

Goal:

> Separate moving foreground objects from static background.

Given:

$$I_t(x,y)$$

and background:

$$B(x,y)$$

Foreground mask:

$$F(x,y) = \begin{cases} 1 & |I_t(x,y) - B(x,y)| > T \\ 0 & otherwise \end{cases}$$

Pipeline

```
Video frame
 ↓
Background model
 ↓
Difference
 ↓
Threshold
 ↓
Foreground mask
```

Applications:

- surveillance
- traffic monitoring
- people counting
- security cameras

43. BACKGROUND MODELING

Instead of using a single background image, we can model pixel distributions.

Example Gaussian model:

$$P(I) = N(\mu, \sigma^2)$$

A pixel is considered foreground if it is sufficiently unlikely under the background model.

More sophisticated models:

- Gaussian mixture models
- adaptive background models

44. OPTICAL FLOW

Optical flow estimates apparent motion of pixels between frames.

Suppose:

$$I(x, y, t)$$

Brightness constancy assumption:

$$I(x, y, t) = I(x + dx, y + dy, t + dt)$$

Using Taylor expansion:

$$I_x dx + I_y dy + I_t dt = 0$$

Divide by dt:

$$\boxed{I_x u + I_y v + I_t = 0}$$

where:

$$u = \frac{dx}{dt}$$

$$v = \frac{dy}{dt}$$

This is the **optical flow constraint equation**.

Problem

One equation contains two unknowns u, v.

This is called the **aperture problem**.

Methods such as Lucas-Kanade add constraints.

45. KLT

Kanade-Lucas-Tomasi tracker

Tracks distinctive feature points between frames.

Typical pipeline:

```
Frame 1
 ↓
Find good features
 ↓
Frame 2
 ↓
Track feature positions
 ↓
Estimate motion
```

KLT commonly uses the Lucas-Kanade optical flow framework.

Good features

Corners are preferred because they provide strong intensity variation in multiple directions.

Applications:

- object tracking
- camera motion estimation
- augmented reality
- video stabilization

46. SPATIO-TEMPORAL ANALYSIS

Spatio-temporal means:

- spatial → x, y
- temporal → t

Instead of considering only:

$$I(x, y)$$

we consider:

$$I(x, y, t)$$

This allows us to analyze changes over time.

Applications:

- action recognition
- motion detection
- activity analysis
- video segmentation

47. DYNAMIC STEREO

Normal stereo estimates depth using multiple cameras.

Dynamic stereo considers **moving scenes/objects**.

Stereo correspondence gives disparity:

$$d = x_L - x_R$$

Depth:

$$\boxed{Z = \frac{fB}{d}}$$

where:

- Z = depth
- f = focal length
- B = baseline
- d = disparity

Numerical example

Suppose:

$$f = 800 \text{ pixels}$$

$$B = 0.1m$$

$$d = 40 \text{ pixels}$$

Then:

$$Z = \frac{800(0.1)}{40}$$

$$\boxed{Z = 2m}$$

48. MOTION PARAMETER ESTIMATION

Motion parameter estimation determines how an object/camera moved.

Examples:

- translation
- rotation
- scaling
- affine transformation
- projective transformation

Translation

$$x' = x + t_x$$

$$y' = y + t_y$$

Affine transformation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

Homogeneous form:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Motion parameters can be estimated using:

- feature correspondences
- least squares
- RANSAC
- optical flow

Canny vs LoG vs DoG

Feature	Canny	LoG	DoG
Principle	Gradient	2nd derivative	Difference of Gaussians
Noise handling	Good	Good after smoothing	Good
Edge localization	Excellent	Good	Moderate
Main use	Edge detection	Edge/blob	Scale-space
Used in SIFT	No	No	**Yes**

Harris vs Hessian-Affine

Harris	Hessian-Affine
Corner detector	Affine-region detector
Uses structure tensor	Uses Hessian
Rotation invariant	Affine adapted
Not scale invariant	Better scale/viewpoint robustness
Good for corners	Good for blobs/local regions

SIFT vs SURF vs HOG vs GLOH

	SIFT	SURF	HOG	GLOH
Main purpose	Local feature	Local feature	Object shape	Local feature
Scale robust	Yes	Yes	Limited	Yes
Rotation robust	Yes	Yes	Limited	Yes
Descriptor	Gradient histogram	Haar responses	Gradient histogram	Gradient + location
Famous use	Matching	Matching	Pedestrian detection	Matching

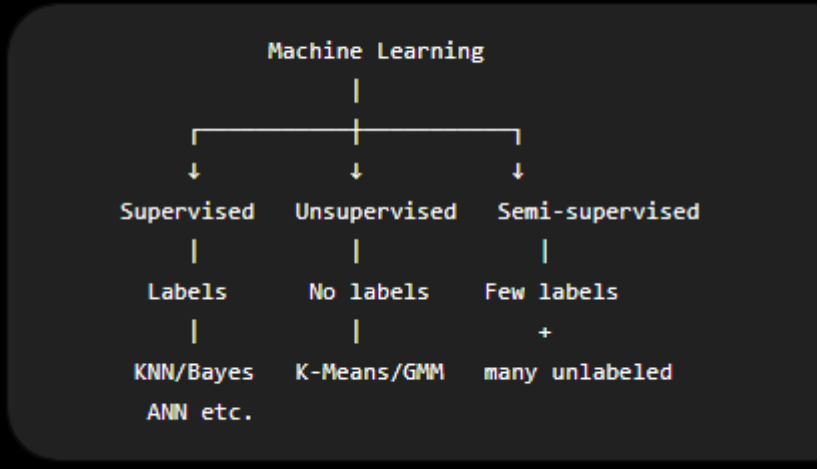
SEGMENTATION COMPARISON

Method	Basic idea	Important property
Region Growing	Similar neighboring pixels	Seed-based
Edge Based	Find boundaries	Boundary-based
Graph Cut	Minimize graph energy	Global optimization
Mean Shift	Find density modes	No predefined K
MRF	Neighbor relationships	Spatial consistency
Texture	Group similar textures	Uses texture features

CLUSTERING COMPARISON

K-Means	K-Medoids	GMM
Centroid	Actual data point	Gaussian distributions
Hard assignment	Hard assignment	Soft assignment
Sensitive to outliers	More robust	Probabilistic
Need K	Need K	Need number of components

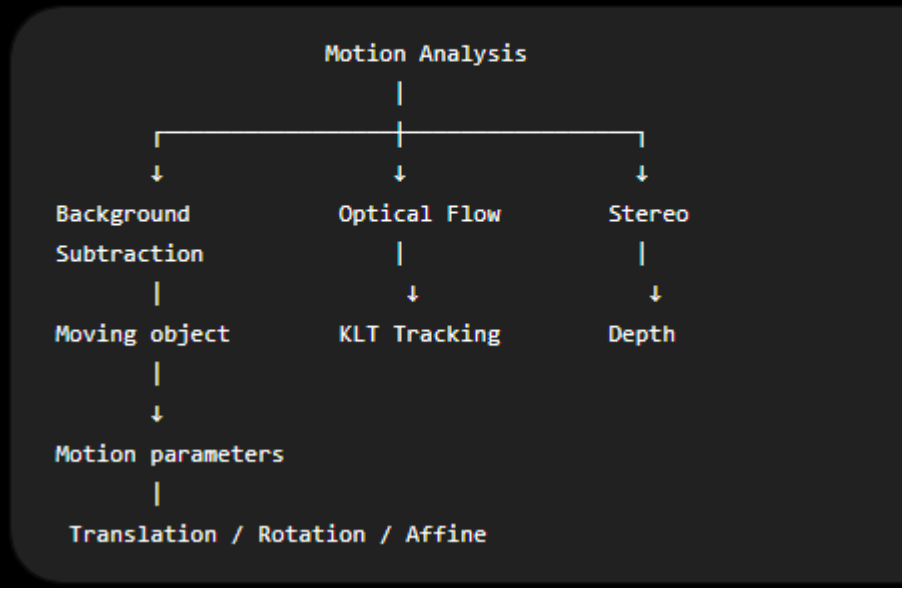
LEARNING TYPES



PCA vs LDA vs ICA

PCA	LDA	ICA
Unsupervised	Supervised	Unsupervised
Max variance	Max class separation	Max independence
Uses covariance	Uses class scatter	Uses statistical independence
Dimensionality reduction	Classification + reduction	Source separation

MOTION ANALYSIS SUMMARY



$$\boxed{\text{IMAGE} \rightarrow \text{FEATURES} \rightarrow \text{SEGMENTATION} \rightarrow \text{PATTERN} \rightarrow \text{OBJECT} \rightarrow \text{MOTION}}$$

Feature extraction asks:

> **What interesting information exists in this image?**

Segmentation asks:

> **Which pixels belong together?**

Pattern analysis asks:

> **What group/class does this feature belong to?**

Object detection asks:

> **What object is present and where?**

Motion analysis asks:

> **How did the object/pixels move?**